AL-KAUSAR
COMPUTER INSTITUTE

ADVANCED COURSE

COURSE 9 OF 9

Full Stack Development

The complete developer path, from zero to a deployed application.

32 weeks	4 × 2 h	256 h	Included
ESTIMATED DURATION	CLASSES PER WEEK	HANDS-ON CLASS TIME	INTERNSHIP + CERTIFICATE

WHAT YOU WILL LEARN

- ✓ Build a complete web application from interface to database.
- ✓ Use React and Next.js the way professional teams do.
- ✓ Design databases and secure APIs.
- ✓ Write tests and ship through a repeatable deployment process.
- ✓ Plan, build and present a product as a small team would.

TOOLS YOU WILL USE

HTML and CSS · JavaScript and TypeScript · React · Next.js · Node.js · PostgreSQL · Tailwind CSS · Git and GitHub

WHO IT IS FOR

Beginners who want one complete path to becoming a developer, and learners who want to build whole products as a freelancer or inside a company.

BEFORE YOU START

None. This is the longest path and it starts from zero.

WHERE IT LEADS

- Full stack developer (junior)
- Frontend or backend developer
- Freelance full stack developer
- Startup product developer

Your course, step by step

11 MODULES · EVERY ONE HAS A HANDS-ON LAB

01 Web foundations — WK 1-4

How the web works · Semantic HTML and forms · CSS: box model, Flexbox and Grid

02 JavaScript — WK 5-8

Variables, functions and scope · Arrays, objects and array methods · The DOM and events

03 TypeScript — WK 9-10

Types, interfaces and generics · Typing functions and data · Narrowing and unions

04 Git, GitHub and Tailwind — WK 11

Commits, branches and pull requests · Working in a team repository · Tailwind: utilities, variants and components

05 React — WK 12-16

Components, props and composition · State and effects · Hooks and custom hooks

06 Next.js — WK 17-20

Routing, layouts and pages · Server and client components · Data fetching and caching

07 Node.js and APIs — WK 21-23

Node and Express · REST design, validation and errors · Middleware

08 PostgreSQL and ORMs — WK 24-26

Tables, relations and keys · Queries, joins and indexes · Migrations and seed data

09 Authentication and security — WK 27-28

Registration, login and sessions · Roles and permissions · Protecting against common attacks

10 Testing and deployment — WK 29

Unit and end-to-end tests · Environment variables and configuration · CI and deployment

11 Capstone project — WK 30-32

Choose and scope a real business application · Design the data, API and screens · Build in weekly milestones

YOU FINISH WITH A REAL PROJECT

A deployed full stack business application, for example a small ERP, with logins and roles, a PostgreSQL database, an admin area, tests, and a final presentation.

INTERNSHIP **Full stack intern.** After the course, work with the team on real tasks, with feedback.

FEE AND NEXT BATCH

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CERTIFICATE

Verify online: kataf.pk/verify